Reward-Channel Separability in Multimodal Chart RLVR: A Cross-Rollout Bonus and a Negative-Sample Caveat

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Abstract

Chart question answering requires a model to parse the visual structure, extract the relevant data, and reason to a final answer. Whether these components respond independently to different training signals, however, is not well understood. We study this with two opposing interventions: **HCPC**, a cross-rollout bonus rewarding consistent extraction and diverse reasoning among correct responses, and Negative Sample Reinforcement (**NSR**; **?**), which removes gradient signal from correct responses entirely.

NSR collapses format compliance from 96.8% to 31.6% on ChartQA with no gain in answer accuracy over GRPO. Per-tag analysis traces the failure to the structural tags reward training had to teach, pointing to reward-signal starvation as the cause. HCPC achieves the best Pass@4 (0.828 vs. 0.816 for GRPO) but degrades out-of-distribution, where reliance on structured extraction becomes a liability. Combining the two, NSR+HCPC recovers GRPO-level accuracy on ChartFC while format compliance stays near zero. The fact that accuracy returns but format does not suggests the two are distinct behavioral channels, and that reward update rules from single-component settings should be re-evaluated before applying them to richer reward landscapes.

1 Introduction

Answering questions about charts requires two sequential operations that are rarely treated as distinct (**????**). A model must first extract the underlying data table from the visual layout, then reason over that table to reach an answer. The first operation has one correct outcome; the second admits many valid paths. Current training pipelines collapse this distinction. The dominant recipe applies GRPO (**?**) with verifiable rewards across chart-type identification, table reconstruction, process conformity, and answer accuracy (**????**), reinforcing all correct rollouts uniformly regardless of which operation they got right or how. The practical cost is gradual: the policy over-commits to whichever solution paths dominate the training distribution, eroding the diversity that supports robustness under distribution shift and coverage at higher K in Pass@K (**?**).

We propose Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus that treats the two operations differently. Among rollouts that get both the table and the answer right, HCPC rewards consistent table extraction and diverse reasoning traces. Consistency is appropriate at the extraction layer, where there is one correct table and reliable recovery is the goal. Diversity is appropriate at the reasoning layer, where multiple paths to the same answer reflect genuine competence. A ground-truth-anchored filter ensures the bonus targets correct-path quality rather than surface agreement among arbitrary rollouts.

To explicitly isolate what HCPC is and is not doing, we introduce a contrastive diagnostic that applies opposing pressure to the same problem. Negative Sample Reinforcement (NSR; **?**) targets the same diversity goal by the opposite route: masking gradient signal on correct rollouts and penalizing only incorrect ones. In single-component math RLVR, this works (**?**). In chart RLVR, it does not. Porting NSR unchanged collapses format compliance, specifically the structured XML tags and table markup that the pipeline depends on, on both in-distribution and out-of-distribution data, with no corresponding gain in answer accuracy. Per-tag analysis traces the failure to exactly the structural tags that reward training had to teach, pointing to reward-signal starvation as the cause. The result is informative precisely because the failure is clean: something about the multimodal, multi-component reward landscape breaks an assumption that held in the simpler setting.

Adding HCPC to NSR recovers competitive

answer accuracy out-of-distribution while format compliance stays near zero. Accuracy returns; format does not. This dissociation is the central finding of the paper. Format compliance and answer accuracy are distinct behavioral channels in chart RLVR, and they can be manipulated independently. The broader implication is not just that NSR fails here, but that reward channels in this setting are separable and should be designed with that separability in mind. A training objective that treats them as one signal will, at best, trade one off against the other; at worst, disrupt both while appearing to recover neither.

Contributions.

1. **HCPC (§3, §4.3).** A cross-rollout bonus applying level-specific pressure to table extraction and reasoning separately. HCPC improves robustness across multi-sample and table-dependence metrics while maintaining parity with the Chart-RVR baseline on single-sample accuracy.

2. **NSR transferability analysis (§4.4).** Porting NSR unchanged into chart RLVR collapses format compliance without accuracy gain. Per-tag analysis identifies reward-signal starvation as the mechanism, localized to structural tags the base model does not emit unprompted.

3. **Separability of format and accuracy (§4.5).** NSR+HCPC recovers competitive out-of-distribution accuracy at near-zero format compliance, providing direct evidence that the two are distinct behavioral channels that respond independently to training interventions. We argue reward components in chart RLVR should be designed with this separability explicit.

2 Background

Chart question answering. Chart question answering maps a chart image I and a question q to an answer a (**??**). Each response decomposes into a trajectory $\tau = (\tau_e, \tau_r)$: τ_e extracts a structured table $\hat{T}$ from the visual layout; τ_r derives a from $\hat{T}$. The two phases differ in correctness structure: τ_e has a unique target T^*, while τ_r admits many valid paths to the same a. We use ChartQA (**?**) (in-distribution) and ChartFC (**?**) (OOD) as our evaluation benchmarks; details in §**??**.

Reinforcement learning with verifiable rewards. RLVR trains a policy π_θ to maximize an expected reward $r(\tau)$ computed programmatically from a verifiable criterion, subject to a KL penalty anchoring the policy to a frozen reference π_{ref}:

$$\mathcal{J}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\cdot|x)}\big[r(\tau)\big] - \beta\, D_{\text{KL}}\big(\pi_\theta(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x)\big).$$

Group Relative Policy Optimization (**?**) approximates the policy gradient by sampling K rollouts per prompt and normalizing rewards within the group; the group mean acts as a self-contained baseline, removing the need for a learned critic (full formula in §4.1). In chart RLVR, structural reward components saturate early while answer accuracy does not (**?**). This asymmetry is the reward-signal mismatch we study.

Chart-RVR backbone. We build on the multi-reward GRPO pipeline of **?**. The structural format reward R_{fmt} (partial-credit, up to 1.4) reaches near-ceiling values well before answer accuracy does. Format adherence is therefore a *learned* behavior reinforced by a dedicated reward, not a byproduct of answer correctness. The exact set of reward components active in our setup is described in §**??**.

PSR/NSR decomposition and transfer challenge. **?** decompose the REINFORCE objective into $\mathcal{L}_{\text{RLVR}} = \mathcal{L}_{\text{PSR}} + \mathcal{L}_{\text{NSR}}$ and show that training on $\mathcal{L}_{\text{NSR}}$ alone matches or surpasses GRPO at Pass@K on math benchmarks. Their rule, however, assumes a single scalar reward. In a multi-reward setting, NSR's skip-correct rule masks rollouts that are well-formed but answer-wrong, removing positive pressure on the structural components the optimizer needs most.

Cross-rollout signals and rollout-set diversity. Group-level training signals condition on $\{o_1, \ldots, o_K\}$ jointly, exploiting within-group variation rather than treating each rollout independently. The inference-time analogue is self-consistency (**?**), which aggregates K sampled answers by majority vote; the group structure there serves calibration, not gradient computation. Pass@K (**?**) evaluates solution-set coverage: a policy that produces a single stereotyped correct path scores high at K=1 but plateaus for $K > 1$, making it a more sensitive diagnostic for diversity than Pass@1 alone. HCPC (§3) is designed to preserve this coverage by conditioning its group-level bonus on ground-truth correctness rather than majority agreement.

3 Method

We build on Chart-RVR (?) as a reward backbone and introduce two interventions: HCPC (Hierarchical Correct-Path Consistency), a group-level bonus rewarding level-specific structure among correct rollouts, and NSR (?), a contrastive gradient-masking strategy that we port to chart RLVR as a diagnostic.

3.1 Preliminaries: Chart-RVR Reward Backbone

We adopt Chart-RVR's per-rollout reward structure, which supervises each output component independently. Rollouts are expected in the format <think><type/> <table/> reasoning</think> <answer/>. Four components are active: R_{fmt} (structural compliance with the required XML template), R_{acc} (answer correctness against y^*), R_{tbl} (table extraction quality assessed by partial structural alignment against T^*), and R_{len} (rationale length and step count). Their sum gives the base reward:

$$R_{\text{base}}(o_i) = R_{\text{fmt}} + R_{\text{acc}} + R_{\text{tbl}} + R_{\text{len}}.$$

Two Chart-RVR components are inactive in our setup: a chart-type reward (R_{type}), which returns zero because the training dataset provides no ground-truth type labels, and a token-count reward (R_{tok}), which we exclude because the base model prepends a native reasoning block that produces duplicate tags. Chart-RVR also defines a process reward measuring similarity to the ground-truth reasoning chain; we disable it as it conflicts with D_{reason}'s diversity objective.

3.2 HCPC: Hierarchical Correct-Path Consistency

HCPC augments R_{base} with a group-level bonus computed exclusively over correct rollouts. The design enforces *level-specific coherence*: consistency where one factually correct answer exists (table extraction) and diversity where multiple paths are valid (reasoning steps). Cross-rollout statistics are conditioned on ground-truth correctness rather than majority agreement (?), preventing shared extraction errors from producing a spuriously high consistency signal.

3.2.1 Correct-Path Filtering

Let $G = \{o_1, \ldots, o_K\}$ be the rollouts for a prompt with ground truth (T^*, y^*). We define the *correct-path subset* $G^+ \subseteq G$ via three sequential gates.

Gate 1 (format). The rollout must be structurally parseable with all required tags and a valid table block.

Gate 2 (table fidelity). The extracted table $\hat{T}_i$ must satisfy $\mathrm{sim}(\hat{T}_i, T^*) \geq \delta$ with δ=0.6, using cosine similarity over MiniLM-L6 embeddings (?) on serialized table strings. This excludes rollouts that reach the correct answer from a misread table.

Gate 3 (answer). The predicted answer $\hat{y}_i$ must match y^*: relative error $\leq 5\%$ for numeric answers, normalized substring match for string answers.

All three gates must pass. An answer-correct rollout with a wrong table has not followed a genuine reasoning path; rewarding its cross-rollout properties would propagate extraction errors into the group signal.

3.2.2 Level-Specific Group Metrics

On G^+ we compute three metrics targeting distinct trajectory levels. C_{type} is the fraction of rollouts in G^+ that predict the modal chart type $\hat{c}_{\text{mode}}$. For table and reasoning levels, let $\langle f \rangle_{G^+}$ denote the mean of $f(i, j)$ over all distinct pairs $i < j$ in G^+:

$$C_{\text{table}} = \big\langle \mathrm{sim}(\hat{T}_i, \hat{T}_j) \big\rangle_{G^+},$$
$$D_{\text{reason}} = 1 - \big\langle \mathrm{sim}(\hat{w}_i, \hat{w}_j) \big\rangle_{G^+}.$$

C_{table} measures pairwise agreement among extracted tables; correct rollouts should converge on the one ground-truth reading. D_{reason} is pairwise dissimilarity among reasoning strings $\hat{w}_i$; high values indicate diverse inferential paths to the same answer. C_{type} and C_{table} enforce convergence where the correct answer is unique; D_{reason} enforces divergence where multiple paths are valid. All similarities use cosine distance over MiniLM-L6 embeddings.

3.2.3 Reward Integration

The consistency component of the group bonus is shared equally among all rollouts in G^+:

$$B_{\text{cons}} = \frac{|G^+|}{K} \big(w_{\text{type}}\, C_{\text{type}} + w_{\text{table}}\, C_{\text{table}} \big).$$

The diversity component is assigned per-rollout via a uniqueness score $u_k = 1 - \bar{s}_k$, where $\bar{s}_k$ is rollout k's mean cosine similarity to all other reasonings in G^+. The per-rollout HCPC reward is then:

$$R_{\text{HCPC}}(o_i) = \begin{cases} B_{\text{cons}} + w_{\text{reason}}\, u_k & \text{if } o_i \in G^+, \\ 0 & \text{otherwise,} \end{cases}$$

with $w_{\text{type}}{=}1.0$, $w_{\text{table}}{=}2.0$, $w_{\text{reason}}{=}1.5$, and $R_{\text{HCPC}}{=}0$ when $|G^+|{<}2$. The total reward is:

$$R_i = R_{\text{base}}(o_i) + R_{\text{HCPC}}(o_i).$$

In practice D_{reason} is small ($\sim$0.05 at $K{=}4$), so the dominant signal is $w_{\text{table}}\, C_{\text{table}}$. HCPC is silent when $|G^+|{<}2$; on ChartQA this occurs 28.8% of the time ($|G^+|{=}0$ on 17.2% of groups, $|G^+|{\geq}3$ on 56.6%), exceeding the naïve Bernoulli bound of 15.7% at $p{=}0.62$ due to within-group visual correlation.

3.3 Diagnostic Baseline: NSR

Negative Sample Reinforcement (**?**) pursues the same diversity goal as HCPC by the opposite route: it removes gradient from correct rollouts entirely and penalizes only incorrect ones. In single-component math RLVR this improves Pass@K (**?**). We port it to chart RLVR unchanged to test whether that finding transfers to a multi-component reward setting.

Masking rule. After GRPO normalizes advantages within a group, we zero out the advantage of any rollout with $R_i \geq \theta$, where $\theta = 0.5\, R_{\max}$. $R_{\max}$ is summed from all defined reward ceilings in the Chart-RVR specification regardless of whether each component is active: $R_{\max}{=}11.0$ for the base configuration and $R_{\max}{=}15.5$ with HCPC ($11.0 + w_{\text{type}} + w_{\text{table}} + w_{\text{reason}}$), giving $\theta{=}5.5$ and $\theta{=}7.75$ respectively. Because two base components are inactive (§**??**), the achievable ceiling is lower; the threshold is therefore conservative, masking only the highest-scoring rollouts. The threshold is fixed rather than group-relative, so it does not shift with the per-step distribution.

NSR+HCPC. When combined, HCPC bonus inflates R_i for rollouts in G^+, making them more likely to be masked by NSR. Rollouts outside G^+ receive no HCPC bonus and are more likely to retain a negative gradient. HCPC membership thus indirectly determines which rollouts drive the policy update under NSR.

4 Experimental Setup

4.1 Data and Training Setup

Training uses the first 1,000 samples of `sanchit97/chart-rvr-grpo-train` (**?**), Chart-RVR's CoT training mixture drawn from ChartQA (**?**), PlotQA, and ChartFC (**?**). Samples are selected sequentially by index. ChartFC samples in the training mixture come from the training split of **?**; the OOD evaluation probe (§**??**) uses the disjoint test split. PlotQA samples are included in training for domain coverage but excluded from evaluation, which targets chart QA and fact-checking directly.

We fine-tune Qwen2.5-VL-3B-Instruct with LoRA ($r{=}8$, $\alpha{=}16$ on W_q, W_v) using AdamW at learning rate 1×10^{-6}, effective batch size 4 (per-device 1, gradient accumulation 4), $K{=}4$ rollouts per prompt at temperature 0.8 and top-$p{=}1.0$, over 2 epochs. No KL penalty is applied ($\beta_{\text{KL}}{=}0$). We evaluate five configurations: BASE (untuned Qwen2.5-VL-3B-Instruct), GRPO, GRPO+HCPC, NSR, and NSR+HCPC. GRPO uses the standard group-normalized advantage $\hat{A}_i = (R_i - \bar{R})/\max(1, s_R)$; NSR additionally zeros out advantages on rollouts with $R_i \geq \theta$ (§**??**).

4.2 Evaluation Protocol

Evaluation samples 4 generations per test prompt at temperature 0.8 and top-$p{=}0.95$ on a 500-sample test slice of each benchmark. **ChartQA** (**?**) is in-distribution; **ChartFC** (**?**) is held out as the OOD probe, differing from ChartQA on task format (fact-checking vs. open-ended QA) and visual style.

We report Pass@1, unbiased Pass@4 (**?**), and format compliance Fmt%. Two additional metrics are central to this paper.

Per-tag emission rate. For each required tag ($\langle$think$\rangle$, $\langle$answer$\rangle$, $\langle$type$\rangle$, $\langle$table$\rangle$, plus proper order), the fraction of rollouts emitting it. This localizes format collapse to specific structural components rather than reporting it in aggregate.

Table-dependence Δ_{tbl}.

$$\Delta_{\text{tbl}} = P(\hat{y} = y^* \mid \text{tbl parsed}) - P(\hat{y} = y^* \mid \text{no tbl}),$$

computed per sample on rollout 0 then aggregated across the test slice. A higher Δ_{tbl} indicates the model's answer correctness depends more strongly on emitting a parseable table. The two arms of the conditional can move for different reasons; we unpack this in §4.3.

5 Results and Analysis

5.1 Main Results

Table 1 reports the five configurations on ChartQA and ChartFC. GRPO training substantially improves both answer accuracy and format compliance over the untuned Base model: Pass@1 rises from 0.518 to 0.630 on ChartQA and Fmt% from 29.8 to 96.8. GRPO+HCPC matches GRPO on Pass@1 (0.622 vs. 0.630) while leading on Pass@4 (0.828 vs. 0.816) and Δ_{tbl} (+0.264 vs. +0.178), consistent with HCPC's design goal of improving rollout-set coverage rather than single-shot accuracy. NSR collapses format compliance to 31.6% on ChartQA and 16.2% on ChartFC with no accuracy gain; the per-tag analysis in §4.4 traces this to reward-signal starvation on structural tags. NSR+HCPC recovers OOD accuracy to 0.632 (statistically tied with GRPO at 0.638; §4.5) while format compliance stays at 17.8%, the format/accuracy dissociation that is the central finding of this paper.

5.2 GRPO+HCPC: Rollout Robustness Without Accuracy Regression

HCPC is designed to strengthen correct-path quality by rewarding consistent extraction and diverse reasoning. The metrics that should reflect this are rollout-set robustness and Pass@K at higher K, not single-shot Pass@1.

Headline Pass@1 on ChartQA is statistically tied: GRPO 0.630 vs. GRPO+HCPC 0.622 (paired bootstrap -0.8 pp, CI $[-3.4, +5.4]$, p=0.77, n_{boot}=10,000). On every other in-distribution metric, the direction favors HCPC: the fraction of test samples where all 4 rollouts are correct is 40.8% for HCPC vs. 38.6% for GRPO; the average correct rate per group is 0.628 vs. 0.620; Pass@4 is 0.828 vs. 0.816; table-dependence is Δ_{tbl}=+0.264 vs. +0.178; format compliance is 98.2% vs. 96.8%. No individual difference reaches significance at n=500 (Pass@4 boot-p=0.48), but the joint direction across six metrics in HCPC's favor is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path quality without trading off single-shot accuracy. Figure 1 shows the Pass@K curves: HCPC matches GRPO at K=1 and pulls

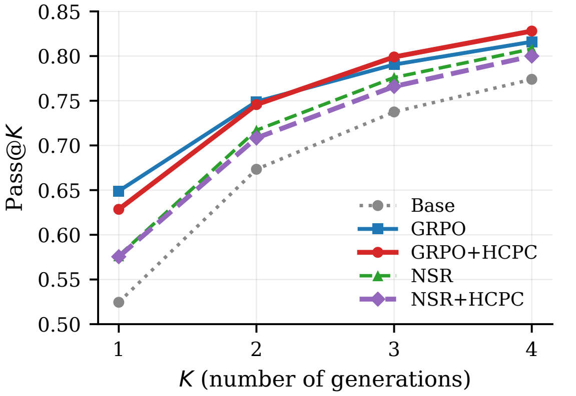

Figure 1: Pass@K on ChartQA (n=500). HCPC's gain over GRPO grows with K: tied at K=1, leading at K=3 and K=4. The direction is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage at higher K. Individual differences are not significant at n=500 (Pass@4 boot-p=0.48).

ahead at $K \geq 3$, the regime where rollout-set coverage matters.

Unpacking Δ_{tbl}: $P(\hat{y} = y^* \mid \text{tbl})$ is 0.639 for GRPO and 0.632 for HCPC (tied; also tied on the common-table intersection of 462 samples, Appendix C), while $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.462 for GRPO and 0.368 for HCPC. The gain therefore reflects stronger table-dependence: HCPC drops 9.4 pp more than GRPO when no table is produced, while matching it when a table is produced. The correct-path bonus makes the extracted table load-bearing in the answer pathway.

GRPO+HCPC OOD limitation. The same property makes GRPO+HCPC weaker than plain GRPO on the OOD ChartFC probe (Pass@1 0.606 vs. 0.638, Pass@4 0.918 vs. 0.922, Δ_{tbl} -0.059 vs. +0.139, format 66.4% vs. 98.8%). The mechanism is plausible from the in-distribution finding: HCPC trains the model to rely on the extracted table, so when table extraction degrades OOD (HCPC's format compliance drops to 66.4% on ChartFC, indicating the extraction pipeline is less reliable on unfamiliar visual styles), the answer pathway degrades with it. The table-dependence that benefits HCPC in-distribution becomes a liability OOD. This is a genuine limitation of standalone GRPO+HCPC, complementary to the NSR-mitigation result in §4.5; we leave a fix (e.g., a chart-style augmentation in training or a softer GT-anchored filter) to future work.

	ChartQA (in-distribution, n=500)				ChartFC (OOD)			
Method	Pass@1	Pass@4	Δ_{tbl}	Fmt%	Pass@1	Pass@4	Δ_{tbl}	Fmt%
BASE	0.518	0.774	-0.031	29.8	–	–	–	–
GRPO	**0.630**	0.816	$+0.178$	96.8	**0.638**	**0.922**	$+0.139$	**98.8**
GRPO+HCPC	0.622	**0.828**	$\mathbf{+0.264}$	**98.2**	0.606	0.918	-0.059	66.4
NSR	0.592	0.808	$+0.082$	31.6	0.590	0.884	-0.059	16.2
NSR+HCPC	0.574	0.800	$+0.147$	43.0	0.632	0.894	$+0.002$	17.8

Table 1: Main results. Δ_{tbl} is the conditional gain in answer correctness from emitting a parseable table; higher means stronger table-dependence. Bold marks column-best per benchmark. Pass@4 uses the unbiased estimator over 4 generations per prompt.

Method	think	answer	type	table	order
BASE	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0

Table 2: Per-tag emission rate (%) on ChartQA. Column headers refer to the `<think>`, `<answer>`, `<type>`, `<table>` tags and the "proper order" check. NSR's collapse is concentrated in the structural tags (think, table) that the base model does not emit unprompted; HCPC under NSR partially recovers table but not think.

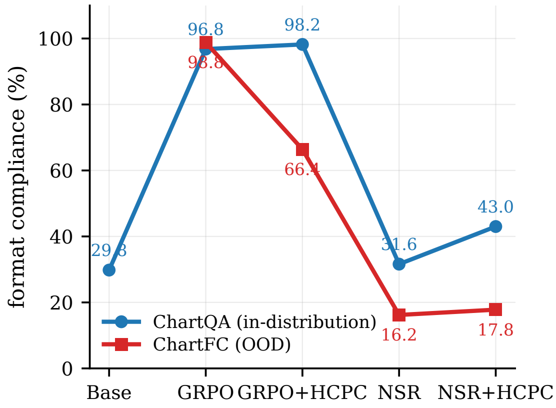

Figure 2: Overall format compliance per method on ChartQA (in-distribution) and ChartFC (OOD). NSR collapses on both; GRPO+HCPC holds ID but degrades OOD; GRPO is the only configuration that keeps format near ceiling under distribution shift.

5.3 NSR Breaks VLM Format

Switching the advantage estimator from GRPO to NSR, with no other changes, drops format compliance from 96.8% to 31.6% on ChartQA and from 98.8% to 16.2% on ChartFC. The Pass@4 gap GRPO vs. NSR on ChartFC is significant: -3.8 pp, CI $[+0.6, +7.2]$, p=0.028 (paired bootstrap), McNemar exact p=0.034.

Table 2 localizes the collapse. NSR keeps the tags the base model already emits ($\langle$type$\rangle$, $\langle$answer$\rangle$, proper order) and loses the ones GRPO has to teach ($\langle$think$\rangle$, $\langle$table$\rangle$). NSR's $\langle$think$\rangle$ rate (57.4%) is essentially the base model's (52.2%) plus noise; $\langle$table$\rangle$ rate (72.2%) falls below the base model's (77.2%).

The mechanism is mechanical. R_{fmt}, the active structural reward in our setup, saturates quickly: emitting the required tags in the correct order is a learned-once behavior that yields near-maximal contribution from that component. With those rewards saturating, the total per-rollout R_i for a well-formed rollout reaches the NSR threshold $\theta = 0.5\,R_{\max}$ regardless of whether the final answer is correct. NSR therefore masks the advantage of *structure-good, answer-wrong* rollouts, leaving only *structure-bad, answer-wrong* rollouts, which receive negative advantage, to drive the policy. This removes the dominant positive gradient on structural-tag emission, and the policy drifts back toward the base model's unstructured output. We expect this failure mode to be absent in **?**'s math experiments because text-math rewards are dominated by a single answer bit; the implicit "format" (a final `\boxed{}` or a number on its own line) is either trivial or emerges from the answer-correctness signal itself. Verifying this empirically is beyond our current scope.

5.4 Format and Accuracy as Separable Channels

On ChartFC, HCPC added to NSR improves over plain NSR on every metric: Pass@1 $0.590 \to 0.632$ (+4.2 pp), Pass@4 $0.884 \to 0.894$ (+1.0 pp), Δ_{tbl} $-0.059 \to +0.002$ (+0.061), and format 16.2% $\to$ 17.8%. The Δ_{tbl} sign flip from negative to zero indicates that table extraction stops actively hurting; the format gain is marginal. Plain NSR is the weakest OOD configuration in Table 1 on every metric, and HCPC closes most of

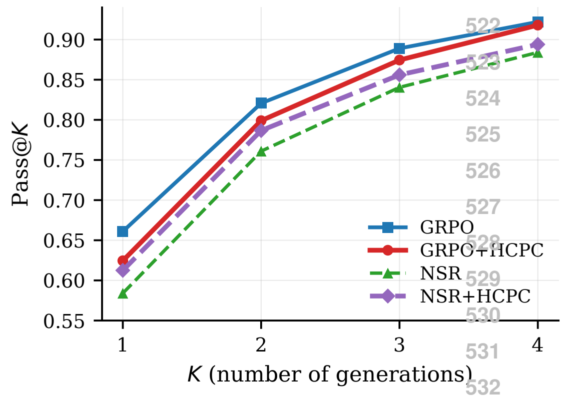

Figure 3: Pass@K on ChartFC (OOD, n=500). Plain NSR (green) sits below all other configurations at every K; adding HCPC (purple) closes most of the gap to GRPO. BASE was not evaluated on ChartFC.

that gap (Figure 2). How the gap is closed (mechanism below) turns out to be more interesting than the gap-closing itself.

Format/accuracy dissociation. The way in which HCPC mitigates NSR's damage is unusual: NSR+HCPC reaches Pass@1 $=$ 0.632, statistically indistinguishable from GRPO (0.638; paired-bootstrap CI $[-0.046, +0.060]$, p=0.85), while format compliance remains collapsed at 17.8% (vs. GRPO's 98.8%). HCPC under NSR recovers GRPO-level OOD answer accuracy without recovering format. This is the cleanest empirical evidence in the paper that format compliance and answer accuracy are separable downstream behaviors in chart RLVR.

Why HCPC behaves so differently under GRPO and NSR. Under GRPO, HCPC behaves as designed: the bonus is added to correct-path rollouts' rewards, those rollouts get higher GRPO advantage $(R_i - \bar{R})/s_R$, and the policy is reinforced toward the correct path. This is the standard positive-gradient interpretation and accounts for the ChartQA results (§4.3).

Under NSR, the same bonus operates through a different channel. HCPC adds its bonus to exactly the correct-path rollouts that NSR then masks out of the gradient, so the bonus does not reach the policy as a direct positive update. But TRL computes the GRPO advantage before the NSR mask is applied, using the group mean $\bar{R}$ over the full set of rollouts. Adding B_{HCPC} to correct rollouts raises $\bar{R}$, and for the wrong rollouts that survive the mask this makes their advantage more negative. The correct-path bonus therefore reshapes the reference baseline that the surviving (wrong) rollouts are penalized against, pushing the policy harder away from wrong-path behavior. The mechanism is *baseline shift, not direct reinforcement*: HCPC under NSR strengthens the negative gradient on wrong rollouts without providing any positive gradient on correct ones. This is consistent with what we observe. Answer-level OOD accuracy recovers because wrong-rollout suppression strengthens, while structured-output emission, which requires positive gradient on correct rollouts that neither NSR nor HCPC-under-NSR can provide, remains collapsed.

On ChartQA, adding HCPC on top of NSR also partially restores format (31.6% $\rightarrow$ 43.0% overall, with the recovery concentrated in ⟨table⟩: 72.2% $\rightarrow$ 83.2%, Table 2). In-distribution, format partially recovers. Out-of-distribution, accuracy recovers fully while format does not. This asymmetry makes the channel-separability claim concrete.

6 Related Work

Chart-reasoning training. Chart-RVR (**?**), a verifiable-reward GRPO pipeline for chart images, is the most closely related to our work, rewarding chart type prediction and table reconstruction alongside answer accuracy. We use its rewards verbatim as R_{base} and add cross-rollout signal; our deltas isolate the effect of the group-level term against the same per-rollout backbone. Other recent chart VLMs improve training data (**???**), reasoning rationales (**???**), or wrap models in agentic pipelines (**????**); these are complementary to our reward-side intervention. Benchmarks beyond ChartQA include ChartX/ChartVLM (**?**) and the style-OOD sets EvoChart (**?**) and ChartQAPro (**?**); we leave style-OOD evaluation as future work.

RLVR and its decomposition. **?** is the only prior work to isolate positive- vs. negative-sample reinforcement; we test their recipe in a multi-component multimodal reward landscape and document the format-collapse failure mode that does not appear in their text-only math setting. **????** study RLVR in other settings but do not address the multi-component-reward asymmetry we identify.

Self-consistency and group-level signals. **?** aggregate by majority vote at inference; **?** balance exploration and exploitation at test time. HCPC differs in being a training-time, group-level reward that conditions on ground-truth correctness rather than majority agreement, and that targets level-specific objectives (consistent extraction, diverse reasoning) rather than a single aggregate.

7 Conclusion

We introduced HCPC, a cross-rollout bonus that preserves correct-path diversity in chart RLVR by conditioning on ground-truth correctness and rewarding consistent extraction together with diverse reasoning. On Qwen2.5-VL-3B, GRPO+HCPC matches the Chart-RVR baseline on Pass@1 and produces consistently more robust correct groups (40.8% all-4-correct vs. 38.6%; Pass@4 0.828 vs. 0.816; Δ_{tbl} +0.264 vs. +0.178). Comparing against Negative-Sample Reinforcement, an alternative diversity-preservation recipe imported from text-only math RLVR, we find that directly porting unchanged NSR collapses format compliance in chart RLVR (96.8% to 31.6%) and significantly hurts OOD Pass@4. This happens because structural rewards saturate quickly enough for well-formed rollouts to cross NSR's correctness threshold regardless of answer quality. Adding HCPC on top of NSR recovers GRPO-level OOD accuracy at near-zero format compliance via a baseline-shift mechanism in the GRPO advantage. This provides empirical evidence that format and answer accuracy are separable behavioral channels of the trained policy. The takeaway for practitioners: advantage-rule asymmetries developed for single-reward source domains can interact destructively with reward components that were absent in the source domain. Therefore, the shape of the multi-component reward mixture deserves to be a first-class consideration when porting RLVR recipes across modalities.

Limitations

We use a single 3B-parameter backbone (Qwen2.5-VL-3B), a 1,000-sample training subset, K=4 rollouts, and a 500-sample test slice per benchmark. Compute constraints prevented us from training on the full Chart-RVR CoT mixture (34,194 samples), evaluating on the full ChartQA and ChartFC test sets, or running EvoChart (**?**) and ChartQAPro (**?**) as style-OOD benchmarks distinct from the joint task-and-style probe ChartFC provides. Scaling to the full training data, full test sets, and a style-only OOD benchmark is our highest-priority follow-up.

A Reproducibility

All training and evaluation code, model checkpoints, and per-sample evaluation outputs (for the 500-sample test slices of ChartQA and ChartFC) are released with the paper. Training uses 4 generations per prompt at temperature 0.8; evaluation uses 4 generations at temperature 0.8 and top-p=0.95. Pass@K uses the unbiased estimator of **?**. Cross-rollout similarities use MiniLM-L6 embeddings. Paired-bootstrap CIs use n_{boot}=10,000 resamples; McNemar tests use the two-sided exact binomial form on the discordant pair.

B Qualitative Example: Format Collapse in Action

Figure-free illustration of the NSR collapse and HCPC partial recovery, on a single ChartQA sample where the question is *"How many categories are there in the chart?"* with ground truth $y^* = 4$. We show rollout 0 from three methods.

GRPO (correct, well-formed).

```
<think><type>pie</type><table>{. . . , "rows": [["Asians","63"
<step-1> The chart is a pie
chart. . . <step-4> Based on the
information provided, there are
four distinct categories. . . </think>
<answer>4</answer>
```

NSR (format-collapsed; arrives at right reasoning but wrong answer surface form).

```
<think>First, I'll identify the type
of chart and the underlying data
table. . . <step-1> The chart shows four
categories: Asians, Whites, Blacks, and
Hispanics. . . . Final answer:
Four
</think> <answer>Four</answer>
```

NSR's rollout contains no `<table>` tag, no `<type>` tag, and emits the reasoning steps inline rather than as structured items. The answer is also expressed in word form ("Four") rather than the numeric form the eval expects ("4"), so the strict-match metric marks it incorrect despite the reasoning being substantively right.

NSR+HCPC (table tag recovered; answer still in word form).

```
<think><type>bar</type><table>{"columns": ["Country","Bar Col
<step-1>From the
chart. . . <step-3>. . . there are
four distinct categories. </think>
<answer>Four</answer>
```

HCPC's group-level bonus on C_{table} pulls NSR's policy back toward emitting a `<table>` tag (table fraction $72.2\% \rightarrow 83.2\%$ on ChartQA, Table 2). The chart type is incorrectly identified (`bar` instead of `pie`), illustrating that HCPC's partial format recovery is not the same as a full quality recovery, and the answer is still "Four" in word form. This is exactly the dissociation pattern of §4.5: format is partly restored, but the answer-surface alignment that drives strict-match accuracy is governed by a separate channel.

C Per-Answer-Type Breakdown

We split ChartQA test samples by ground-truth answer type. Pass@1 differences between GRPO and GRPO+HCPC are within exact-McNemar noise on numeric (n=302, -1.7 pp, p=0.63), boolean (n=75, $+4.0$ pp, p=0.71), and string (n=114, -6.1 pp, p=0.30). HCPC does not provide a category-specific accuracy advantage at this scale.

D Δ_{tbl} on the Common-Table Subset

To rule out the confound that Δ_{tbl} is computed on different denominators per method, we recompute it on the intersection of samples where multiple methods produce a parseable table. On the 462 ChartQA samples where both GRPO and GRPO+HCPC produce parseable tables, $P(\hat{y} = y^* \mid \text{tbl})$ is 0.643 vs. 0.636. The Δ_{tbl} gap in Table 1 therefore arises entirely from the $P(\hat{y} = y^* \mid \text{no tbl})$ arm (0.462 vs. 0.368). We interpret this as evidence that HCPC induces stronger table-dependence, not stronger table-conditioned reasoning accuracy.